

Photograph the Moon

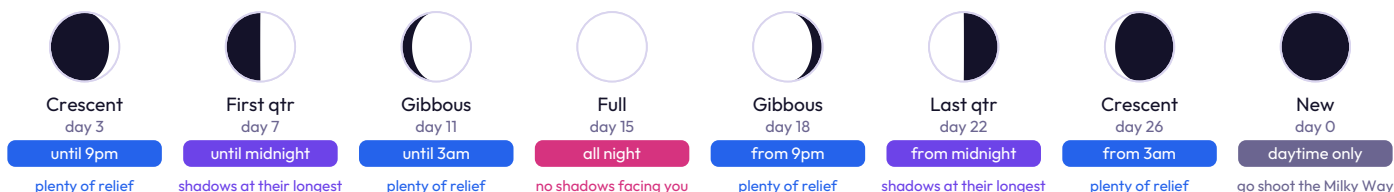
Print it, fold it, take it out with you.

Fragments of the Universe
bhapstar.com

The whole technique, in one paragraph

The Moon is a sunlit rock, so expose it like something in daylight, not like the night sky. Set the camera to manual, f/8 to f/11, ISO 100 to 200, and a shutter of one over the ISO number. Nothing needs to track, because at those speeds the sky has not moved. Then pick a night that is not full: all the detail lives along the terminator, the curved line between the lit and unlit halves, where the shadows are long.

IS IT UP TONIGHT, AND IS TONIGHT WORTH IT



The Moon rises about fifty minutes later each day, so it works its way round the clock once a month. Times above are nominal and move with your latitude and the season, but the order never changes: the first half of the month is an evening Moon, the second half is a small-hours Moon, and for three or four nights around new there is nothing to photograph at all. Those are the nights the rest of astrophotography waits for.

CAMERA ON A TRIPOD, ALL MANUAL

Mode	Manual	Every auto mode meters the black sky and blows the Moon out
Aperture	f/8 to f/11	Most lenses are sharpest a stop or two down from wide open
ISO	100 to 200	There is plenty of light. Low ISO keeps the file clean
Shutter	One over the ISO	ISO 200 means 1/200s. The Looney 11 rule. Then adjust by eye
Focus	Manual, magnified	Put a crater edge or the terminator centre screen and turn until the shadows snap. Re-check after half an hour as the lens cools
File	RAW	Keeps the detail in the bright areas that a JPEG throws away
Drive	Burst of 20 or more	The air is moving. Some frames caught it holding still
Stabilisation	Off	On a tripod it can add the shake it exists to remove
Tracking	Not needed	At 1/200s the sky has not moved. A wall will do

A crescent wants a little more light than a full Moon, because you are seeing the surface at a low angle. Start at the numbers above and open up half a stop at a time until the terminator stops looking empty.

HOW MUCH OF THE FRAME THE MOON FILLS

100mm 4% a dot	200mm 8% a small disc	400mm 15% croppable	800mm 30% a real picture	1600mm 60% craters everywhere	2700mm 100% edge to edge
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Measured on a full-frame sensor. A crop body shows the Moon about half again as large at the same focal length, and a small astronomy camera larger still, because both are cropping into the middle of the same picture.

WHAT LIMITS YOU IS THE AIR

Wait until it is high

Near the horizon you are looking through several times more air. Thirty degrees up is a sensible minimum. Higher is better.

Never over a hot roof

A road, a rooftop or an air conditioning unit that baked all day pours heat upwards all evening. In a city this ruins more nights than light pollution does.

Stack the sharpest

Shoot a burst or a short video, throw away the soft frames and average the rest. Siril does it free. That is how sharp lunar pictures are made.

BEFORE YOU PRESS ANYTHING

Phase not full | Altitude over 30 degrees | Exposure manual, way down | Focus magnified, by hand | Frames a burst, not one

The full guide, with examples

Full walkthrough with example images, the phase timetable in detail, how to tell in ten seconds whether the Moon is up tonight, and what to photograph on the few nights each month when it is not.

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